

Esha Sood

571-549-9989 | applicationsbyesha@gmail.com | <https://www.linkedin.com/in/esha-sood>

EDUCATION

Arizona State University, Tempe, AZ

Master of Science in Computer Science

GPA: 3.93/4.00

Aug. 2022 – Dec. 2023

Maharaja Sayajirao University of Baroda, Vadodara, India

Bachelor of Engineering in Electronics(Gold Medalist)

GPA: 4.00/4.00

Jul. 2018 – May 2022

EXPERIENCE

Engineering Analyst

Goldman Sachs

Feb 2024 – Present

Salt Lake City, UT

- Automated manual trade allocations by designing and implementing a streamlined workflow, reducing operational load and increasing efficiency by 20%
- Utilized technologies such as Java, Kafka, and Kubernetes to migrate legacy systems written in proprietary languages to decoupled and high-performing Java-based architectures, enhancing efficiency and scalability

Engineering Summer Analyst

Goldman Sachs

Jun. 2023 – Aug. 2023

Salt Lake City, UT

- Designed and implemented a high-performance message storage system for an application handling 15 million daily messages, utilizing an asynchronous event logger with Logback to efficiently write data to files and Fluent Bit to upload them to Google Cloud Platform
- Built a centralized, scalable storage solution enabling efficient search and filtering of high-volume messages, improving debugging efficiency by 40%

PROJECTS

Fraud Detection with Graph Databases and ML | *ML Algorithms, Neo4j, Scikit-learn, D3.js*

Fall 2023

- Built a fraud detection framework using graph-based features and machine learning models (Random Forest, Logistic Regression, SVM) on the BankSim dataset, improving detection accuracy by 50%
- Integrated Neo4j for graph modeling, applied algorithms like PageRank, and developed a dynamic UI with D3.js and Matplotlib to visualize fraud patterns and model performance

Soccer Team Performance Prediction | *ML Algorithms, Scikit-learn, Matplotlib, Pandas, Numpy*

Fall 2023

- Built a machine learning model predicting English Premier League outcomes with 78.7% accuracy using a Random Forest Classifier and advanced feature engineering on team metrics, historical data, and player statistics
- Compared six models (e.g., XGBoost, AdaBoost) and optimized features like team form and shot conversion rates, boosting F1 scores from 62% to 69.9%

KubeFlow Data Processing Framework | *Kubernetes, Kafka, Neo4j, Docker, Helm, Python*

Spring 2023

- Built a Kubernetes-based data pipeline using Kafka and Neo4j for real-time streaming, storage, and graph analytics on large datasets, integrating advanced algorithms like PageRank and BFS for actionable insights
- Utilized tools like Minikube, Helm, and Kafka Connect to enable seamless communication and scalable, decoupled data processing architectures

Serendip System | *D3.js(JavaScript), HTML/CSS, Python(Scikit-learn)*

Fall 2022

- Developed a topic modeling and visualization system using Latent Dirichlet Allocation (LDA) to analyze a poetry dataset from diverse eras and themes, offering insights through interactive views like Corpus, Text and Rank Viewer
- Implemented interactive visualizations with D3.js, scikit-learn, and HTML/CSS, and added a Word Viewer feature to highlight frequently used words and their contextual significance across the poetry dataset

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, Embedded C

Technologies: Kafka, Kubernetes, Google Cloud Platform, Fluent-bit, Neo4j, Helm

Developer Tools: Git, GitHub, Docker, IntelliJ IDEA, PyCharm, VS Code, Visual Studio, Eclipse, MATLAB

Web Development and Frameworks: ReactJS, Node.js, Spring Boot, Bootstrap, D3.js, JUnit

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib